# Lect#3:
# Project Integration Management(PIM)

- What is PIM
- Project Plan development
- Project Plan Execution
- Overall Change Control
- Need for top Management Commitment

# What is Project Integration Management(PIM)

- PIM includes processes involved in coordinating all 8 project mgmt knowledge areas throughout a project's life cycle.
- It ensure that all elements of a project come together at the right times to complete a project successfully.
- PIM, key to overall project success.
- Project integration management is not the same thing as software integration

# 3 Main Processes in PIM

1. **Project Plan Development: taking the results of other planning processes and putting them into a consistent, coherent document—the project plan**
2. **Project Plan Execution: carrying out the project plan**
3. **Overall Change Control: coordinating changes across the entire project**

# Project Integration Management Overview

**PROJECT INTEGRATION MANAGEMENT**

**4.1 Project Plan Development**

**.1 Inputs**
- .1 Other planning outputs
- .2 Historical information
- .3 Organizational policies
- .4 Constraints
- .5 Assumptions

**.2 Tools and Techniques**
- .1 Project planning methodology
- .2 Stakeholder skills and knowledge
- .3 Project management information system (PMIS)
- .4 Earned value management (EVM)

**.3 Outputs**
- .1 Project plan
- .2 Supporting detail

**4.2 Project Plan Execution**

**.1 Inputs**
- .1 Project plan
- .2 Supporting detail
- .3 Organizational policies
- .4 Preventive action
- .5 Corrective action

**.2 Tools and Techniques**
- .1 General management skills
- .2 Product skills and knowlede
- .3 Work authorization system
- .4 Status review meetings
- .5 Project management information system
- .6 Organizational procedures

**.3 Outputs**
- .1 Work results
- .2 Change requests

**4.3 Integrated Change Control**

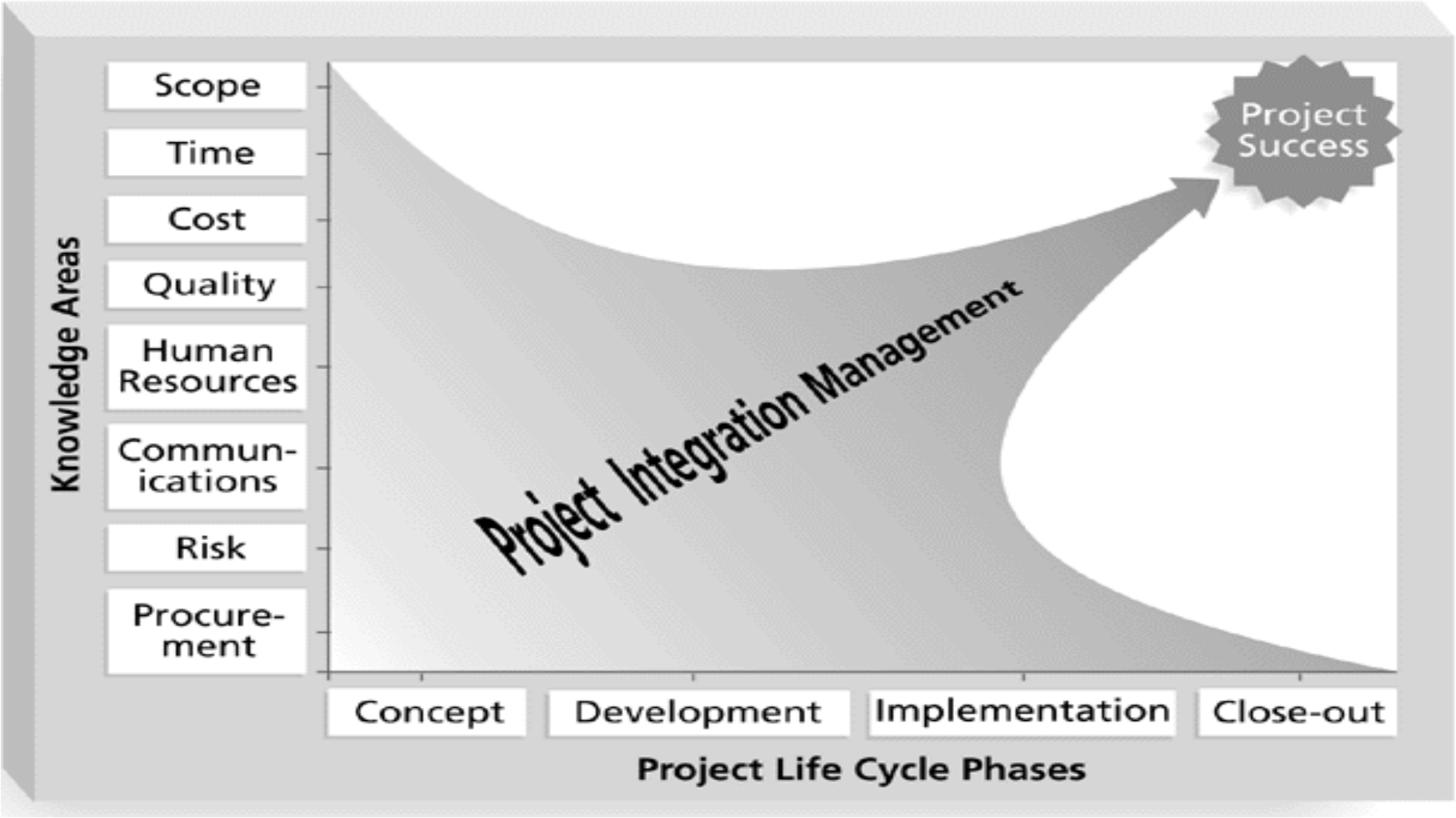
**.1 Inputs**
- .1 Project plan
- .2 Performance reports
- .3 Change requests

**.2 Tools and Techniques**
- .1 Change control system
- .2 Configuration management
- .3 Performance measurement
- .4 Additional planning
- .5 Project management Information system

**.3 Outputs**
- .1 Project plan updates
- .2 Corrective action
- .3 Lessons learned

*PMBOK Guide*, 2000, 42.

# Figure 3-1. Framework for Project Integration Management



# PIM

- Someone must take responsibility
  - for coordinating all of the people, plans, and work required to complete a project.
  - to make the final decisions when there are conflicts among project goals or people involved
  - must communicate key project information to senior mgmt.
- This someone is Project Manager

# Interface Management

- Good PIM is critical to provide stakeholder satisfaction & its includes interface management
- Interface management involves identifying and managing the points of interaction between various elements of the project. As number of people involve increase, interface increase.
- Thus, project managers must establish and maintain good communication and relationships across organizational interfaces

# 1. Project Plan Development

- A project plan is a document used to coordinate all project planning documents
- Its main purpose is to *guide project execution & control*
- Project plans assist the project manager in leading the project team and assessing project status

# Attributes of Project Plans

- Just as projects are unique, so are project plans
  - Plans should be dynamic
  - Plans should be flexible
  - Plans should be updated as changes occur
  - Plans should first and foremost guide project execution

# Common Elements of a Project Plan

- Introduction or overview of the project
- Description of how the project is organized
- Management and technical processes used on the project
- Work to be done, schedule, and budget information

# Table 3-1. Sample Outline for a Software Project Management Plan (SPMP)

| Project Management Plan Sections | | | | | |
|---|---|---|---|---|---|
| | **Introduction** | **Project Organization** | **Managerial Process** | **Technical Process** | **Work Packages, Schedule, and Budget** |
| Section Topics | Project overview; project deliverables; evolution of the SPMP; reference materials; definitions and acronyms | Process model; organizational structure; organizational boundaries and interfaces; project responsibilities | Management objectives and priorities; assumptions, dependencies, and constraints; risk management; monitoring and controlling mechanisms; and staffing plan | Methods, tools, and techniques; software documentation; and project support functions | Work packages; Dependencies; resource requirements; budget and resource allocation; and schedule |

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# Stakeholder Analysis

- Because the ultimate goal of PM is to meet/exceed stakeholder needs & expectation, important to have stakeholder analysis as part of project planning
- A stakeholder analysis documents important (often sensitive) information about stakeholders such as
  - stakeholders' names and organizations
  - roles on the project
  - unique facts about stakeholders
  - level of influence and interest in the project
  - suggestions for managing relationships

# Table 3-2. Sample Stakeholder Analysis

| Key Stakeholders | | | | | |
|---|---|---|---|---|---|
| | **Ahmed** | **Susan** | **Erik** | **Mark** | **David** |
| Organization | Internal senior management | Project team | Project team | Hardware vendor | Project manager for other internal project |
| Role on project | Sponsor of project and one of the company's founders | DNA sequencing expert | Lead programmer | Supplies some instrument hardware | Competing for company resources |
| Unique facts | Demanding, likes details, business focus, Stanford MBA | Very smart, Ph.D. in biology, easy to work with, has a toddler | Best programmer I know, weird sense of humor | Start-up company, he knows we can make him rich if this works | Nice guy, one of oldest people at company, has 3 kids in college |
| Level of interest | Very high | Very high | High | Very high | Low to medium |
| Level of influence | Very high; can call the shots | Subject matter expert; critical to success | High; hard to replace | Low; other vendors available | Low to medium |
| Suggestions on managing relationship | Keep informed, let him lead conversations ,do as he says and quickly | Make sure she reviews specifications and leads testing; can do some work from home | Keep him happy so he stays; emphasize stock options; likes Mexican food | Give him enough lead time to deliver hardware | He knows his project takes a back seat to this one, but I can learn from him |

# 2. Project Plan Execution

- Project plan execution involves managing and performing the work described in the project plan
- The majority of time and money is usually spent on execution
- The application area of the project directly affects project execution because the products of the project are produced during execution

# Important Skills for Project Execution

- General management skills like leadership, communication, and political skills are essential
- Product skills and knowledge
- Use of specialized tools and techniques

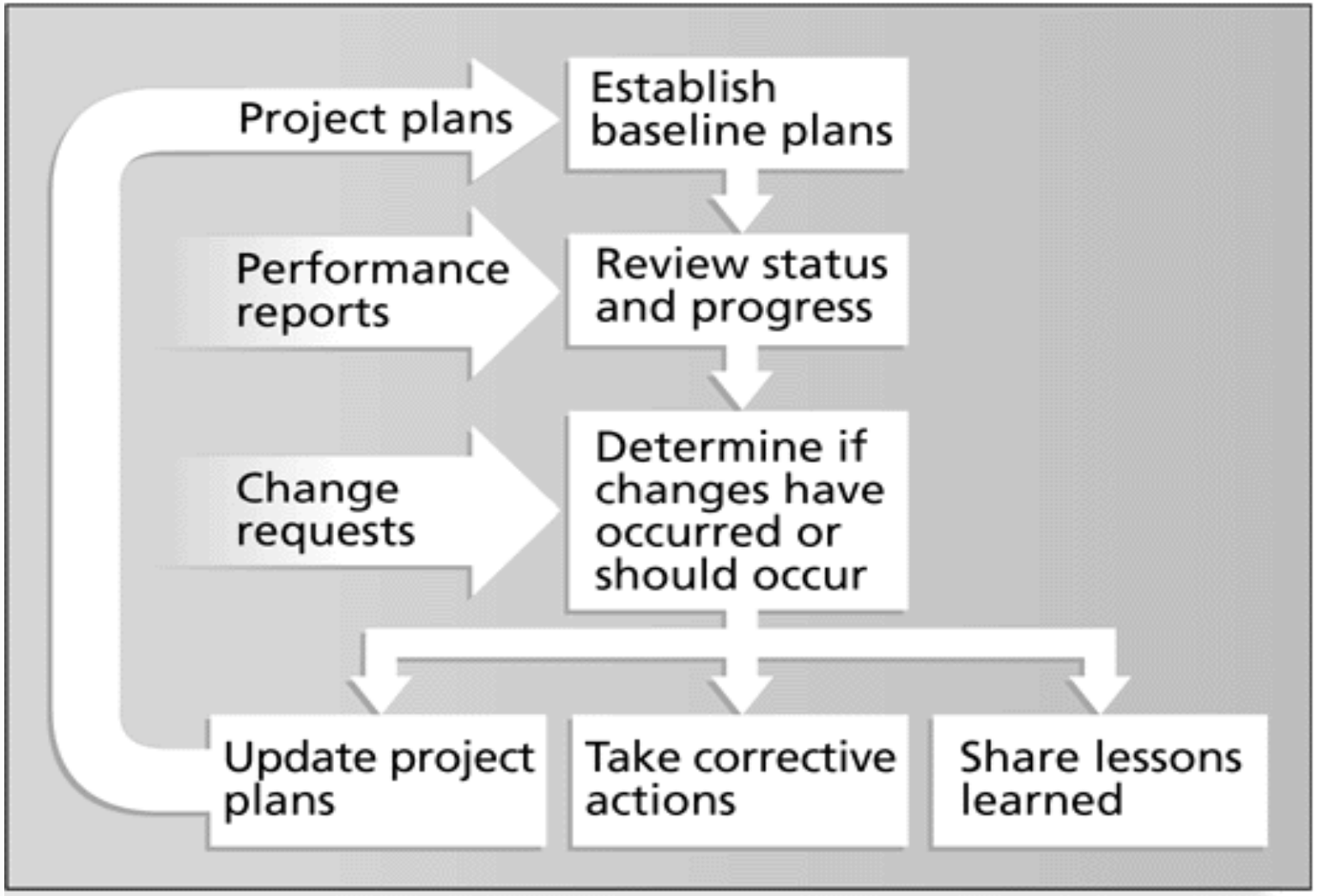
# Tools and Techniques for Project Execution

- **Work Authorization System: provides a method for ensuring that qualified people do work at the right time and in the proper sequence**
- **Status Review Meetings: regularly scheduled meetings used to exchange project information**
- **Project Management Software: special software to assist in managing projects**

# 3. Overall Change Control (OCC)

- OCC involves identifying, evaluating, and managing changes throughout the project life cycle
- Three main objectives of change control:
  - Influence the factors that create changes to ensure they are beneficial
  - Determine that a change has occurred
  - Manage actual changes when and as they occur

# Figure 3-2. Overall Change Control Process



# Table 3-3. Suggestions for Managing Overall Change Control

- View project management as a process of constant communications and negotiations
- Plan for change
- Establish a formal change control system, including a Change Control Board (CCB)
- Use good configuration management
- Define procedures for making timely decisions on smaller changes
- Use written and oral performance reports to help identify and manage change
- Use project management and other software to help manage and communicate changes

# Need for Top Management Commitment

- Several studies cite top management commitment as one of the key factors associated with project success
- Top management can help project managers secure adequate resources, get approval for unique project needs in a timely manner, receive cooperation from people throughout the organization, and learn how to be better leaders

# Need for Organizational Commitment to Information Technology (IT)

- If the organization has a negative attitude toward IT, it will be difficult for an IT project to succeed
- Having a Chief Information Officer (CIO) at a high level in the organization helps IT projects
- Assigning non-IT people to IT projects also encourages more commitment

# Need for Organizational Standards

- Standards and guidelines help project managers be more effective
- Senior management can encourage
  - the use of standard forms and software for project management
  - the development and use of guidelines for writing project plans or providing status information
  - the creation of a project management office or center of excellence